Every term of OEIS A393856 is congruent to 1 modulo 5

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Abstract

OEIS A393856 is defined by the functional equation $A\big(x - xA(4x)/4\big) = x$ for its ordinary generating function, and carries a conjecture of P. D. Hanna that every term is congruent to 1 modulo 5. It is true, and the reason is that $4 \equiv -1$ modulo 5. The functional equation determines the coefficients one at a time, so it determines them modulo 5 as well; and modulo 5 the series $x/(1-x)$, all of whose coefficients are 1, satisfies the equation exactly. Uniqueness then forces $A(x) \equiv x/(1-x)$, which is the conjecture. The argument is an identity between rational functions over $\mathbb{Z}/5\mathbb{Z}$ and involves no estimation and no truncation.

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1 The sequence and the conjecture

OEIS A393856 is "G.f. A(x) satisfies: A(x - x*A(4*x)/4) = x.". It has offset 1 and begins

$$1, 1, 6, 121, 8466, 2227706, \ \dots$$

So, writing $A(x) = \sum_{n\geq 1} a(n)x^n$, the entry defines A by

$$A\left(x - \frac{x\,A(4x)}{4}\right) \;=\; x. \tag{1}$$

This is the only input taken from the entry.

The entry separately carries the following comment:

> Conjecture: a(n) == 1 (mod 5) for n >= 1.

As of the "Last modified" line on the live entry (Mar 12 2026 10:39:38, revision 14) the statement is still recorded as a conjecture, and nothing on the entry records it as settled either way.

Written out, the conjecture asserts

$$a(n) \;\equiv\; 1 \pmod 5 \qquad \text{for every } n \geq 1. \tag{2}$$

2 The equation determines the sequence

It is convenient to clear the division. Since A has no constant term,

$$B(x) \;:=\; x - \frac{x\,A(4x)}{4} \;=\; x - \sum_{n\geq 1} a(n)\,4^{n-1}x^{n+1}, \tag{3}$$

so B has integer coefficients even though (1) appears to divide by 4.

Lemma 1. *There is exactly one sequence of integers $a(1), a(2), \ldots$ satisfying (1), and it is given by $a(1) = 1$ together with*

$$a(N) \;=\; -\sum_{j=1}^{N-1} a(j)\,\big[x^N\big]B(x)^j \qquad (N \geq 2), \tag{4}$$

in which the right-hand side depends only on $a(1), \ldots, a(N-1)$.

Proof. $B(x) = x + O(x^2)$, so B has a compositional inverse and $\big[x^j\big]B^j = 1$ while $\big[x^N\big]B^j = 0$ for $N < j$. Expanding (1),

$$x \;=\; A\big(B(x)\big) \;=\; \sum_{j\geq 1} a(j)\,B(x)^j.$$

Comparing coefficients of x^1 gives $a(1) = 1$. For $N \geq 2$, comparing coefficients of x^N gives

$$0 \;=\; \sum_{j=1}^{N} a(j)\,\big[x^N\big]B^j \;=\; a(N) + \sum_{j=1}^{N-1} a(j)\,\big[x^N\big]B^j,$$

since the $j = N$ term contributes $a(N) \cdot 1$. This is (4). For the dependence claim: by (3), $\big[x^i\big]B$ involves only $a(i-1)$, so $\big[x^N\big]B^j$ involves only $a(1), \ldots, a(N-1)$ for every $j \geq 1$. Hence (4) determines $a(N)$ from its predecessors, and induction gives both existence and uniqueness. $\square$

Corollary 2. *For any integer $k \geq 2$, the reductions $a(n) \bmod k$ are determined by the same recursion read in $\mathbb{Z}/k\mathbb{Z}$. Consequently, if $\bar{A}(x) \in (\mathbb{Z}/k\mathbb{Z})[[x]]$ has $\bar{A}(x) = x + O(x^2)$ and satisfies (1) with coefficients read modulo k, then $a(n) \equiv \big[x^n\big]\bar{A}(x) \pmod{k}$ for every n.*

Proof. Every operation in (3) and (4) is a sum or product of integers, so reduction modulo k is a ring homomorphism carrying the integer recursion to the same recursion over $\mathbb{Z}/k\mathbb{Z}$. That recursion has a unique solution, by the argument of Lemma 1 verbatim; both $a \bmod k$ and $\bar{A}$ solve it. $\square$

3 The reduction modulo 5

Everything now happens in $(\mathbb{Z}/5\mathbb{Z})[[x]]$. The one fact that drives the proof is

$$4 \;\equiv\; -1 \pmod{5}. \tag{5}$$

Theorem 3. $a(n) \equiv 1 \pmod{5}$ *for every $n \geq 1$. This is the conjecture of Section 1.*

Proof. Let $\bar{A}(x) = \dfrac{x}{1-x} = \sum_{n\geq 1} x^n$ in $(\mathbb{Z}/5\mathbb{Z})[[x]]$, so all its coefficients are 1 and $\bar{A}(x) = x + O(x^2)$. By Corollary 2 it suffices to check that $\bar{A}$ satisfies (1) modulo 5.

Form $\bar{B}$ from $\bar{A}$ by (3). Its coefficients are $\big[x^{n+1}\big]\bar{B} = -4^{n-1}$, and by (5) $4^{n-1} \equiv (-1)^{n-1}$, so

$$\bar{B}(x) \;=\; x - \sum_{n\geq 1}(-1)^{n-1}x^{n+1} \;=\; x - x\sum_{n\geq 1}(-1)^{n-1}x^n \;=\; x - x \cdot \frac{x}{1+x} \;=\; \frac{x}{1+x}.$$

Composing,

$$\bar{A}\big(\bar{B}(x)\big) \;=\; \frac{\dfrac{x}{1+x}}{1 - \dfrac{x}{1+x}} \;=\; \frac{\dfrac{x}{1+x}}{\dfrac{(1+x)-x}{1+x}} \;=\; \frac{x}{1+x} \cdot \frac{1+x}{1} \;=\; x.$$

Both compositions are legitimate because $\bar{B}(x) = x + O(x^2)$ has zero constant term. So $\bar{A}$ satisfies (1) modulo 5, and Corollary 2 gives $a(n) \equiv \big[x^n\big]\bar{A} = 1 \pmod{5}$ for every $n \geq 1$. $\square$

Remark 4. The proof used nothing about 4 beyond (5). The same argument shows that for every integer $m \geq 2$, the sequence defined by $A\big(x - xA(mx)/m\big) = x$ has all of its terms congruent to 1 modulo $m + 1$. Note also where the hypothesis bites: modulo k the quantity 4^{n-1} must be $(-1)^{n-1}$ for the telescoping above, and that is exactly (5).

4 Verification

Three checks, each independent of the algebra above.

First, the recursion (4) was run in exact integer arithmetic and its output compared against the entry's DATA at every one of the 15 published indices; the values agree exactly. A recursion that reproduced only some of them would mean the functional equation had been transcribed wrongly.

Second, the same run was continued to $n = 24$ and every term reduced modulo 5; all 24 residues are 1, as Theorem 3 requires.

Third, the composition $\bar{A}\big(\bar{B}(x)\big)$ of the proof was carried out independently as a formal power series over $\mathbb{Z}/5\mathbb{Z}$, coefficient by coefficient to order 30, starting from $\bar{a}(n) = 1$ rather than from the closed form; the result is x exactly, and the computed $\bar{B}$ agrees with $x/(1+x)$ at every one of those orders.

References

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